An Insidious Takeover by Microplastics

Abstract

Microplastics are the hidden side of global pollutants. Through this project, we have created data-driven storytelling and innovative visualizations to expose how microplastics insidiously embed themselves into our ecosystems and even within the human body. We collected effective and informative data from various sources, including WHO, IUCN, EPA, UNEP, etc. By analyzing these datasets, we arrived at findings that depict how microplastics find their way back into the human body and the various health hazards they cause. Through interactive visual idioms like tree maps, stream graphs, bar charts, donut charts, geographic scatter maps, heat maps, radar charts, etc., we aim to raise awareness of microplastic pollution and offer a visually compelling narrative that bridges scientific insight with public understanding.

1 Problem Statement

How do microplastics travel from everyday products to our bodies and what harm do they cause? This project reveals their invisible journey through interactive maps and animations. We show where microplastics come from (like synthetic textiles or packaging), how they spread through rivers and oceans, where they accumulate in human organs, and the proven health risks they pose. By making complex science accessible, we empower individuals and policymakers to take action.

2 Background

Plastic, once hailed as a revolutionary material, has become a pervasive environmental threat due to unchecked production and inadequate disposal systems. Microplastics, defined as plastic particles less than 5mm in size, are particularly dangerous because of their invisible presence and persistent nature. They originate from sources like industrial waste, synthetic fabrics, and packaging materials, ultimately making their way into rivers, oceans, and food systems. Recent scientific studies reveal alarming evidence: microplastics have been detected in human bloodstreams, vital organs like the liver and brain, and even placental tissue. Despite the severity of this issue, public awareness remains limited. Our project aims to bridge this gap by curating a sequence of interactive visualizations that trace the path of microplastics across global geographies and human physiology. Using a combination of scatter geo plots, radar charts, scrollytelling techniques, and choropleth maps, our team provides a layered narrative on microplastic infiltration and its potential consequences.

3 Datasets

- Microplastics in Different Areas of the Brain (2016 vs 2024): self-derived
- World Plastic Exports: https://www.kaggle.com/datasets/sohamgade/plastic-datasets
- Country-wise Plastic Pollution: https://ourworldindata.org/plastic-pollution

- Dietary Microplastic Intake (2018): https://pubs.acs.org/doi/10.1021/acs.est.4c00010
- Marine Animal Ingestion Analysis: self-generated
- Human Body Burden: self-derived
- Top Plastic-Producing Companies: https://cdn.minderoo.org/content/uploads/2021/05/27094234/20211105-Plastic-Waste-Makers-Index.pdf

4 Narrative

We wanted to tell a story about how plastics, something we use every day without thinking, doesn't just disappear after we throw it away. It travels around the globe and ultimately finds its way back to us.

Our story begins at the source. Microplastics come from surprising places—tiny particles from worn-out tires, road paint, and even the clothes we wear. These tiny fragments flow through drains and rivers, eventually reaching our oceans. To show this movement, we used a streamgraph—one that visualizes how plastic flows from different origins into the sea, country by country.

From there, we zoom out. Using a global map dotted with interactive points, we show where plastic concentrations are highest in the world's oceans. Clicking on these dots reveals the makeup of these ocean garbage patches—tiny bits of bottles, bags, and packaging, each with a story of its own.

Then we turn to rivers—the arteries carrying this pollution. An interactive chart lets you explore which rivers carry the most plastic, and what kinds. The data becomes more unsettling as we move to marine life. Another chart shows how birds and fish are ingesting these plastics, sometimes mistaking them for food.

But it doesn't stop with nature—it comes back to us. We visualized how microplastics end up in our meals and drinking water. Another chart shows how these plastics embed themselves in human organs. Most haunting of all, we share research on microplastics found in the human brain—and how their presence has increased over time.

And finally, we ask: who is responsible? In a bold, experimental chart, each company's contribution to global plastic production is shown as a vibrating line. The more plastic they produce, the more violently the line shakes—a visual metaphor for the damage they leave behind.

We designed this experience to make the invisible visible. By connecting plastic's long journey—from the tires on our roads to the food on our plates—we hope to shift the conversation. Because this isn't just about pollution. It's about us.

5 Innovative Visualizations

- **Streamgraph**: A streamgraph to show the flow of microplastic sources (e.g., tire dust, road markings) into water systems.
- **Bar and Donut Charts**: Sector and type comparisons, focusing on textiles and fibers.
- **Geographic Scatter Map**: A combinatorial interactive world map to display varying plastic concentrations across oceans, with clickable circular markers triggering pop-up breakdowns of garbage patch contents.
- **Human-Body Diagram**:Designed an interactive human-like SVG where each brain region is clickable, displaying specific impacts of microplastics on that area, allowing users to trace how accumulation progresses and affects brain function.
- **Network Diagram**: Developed a temporal chart to visualize plastic accumulation across different brain regions over time, incorporating Bollinger Bands to highlight trends and fluctuations in microplastic concentration.
- **Vibrating Line Chart**: A vibrating line chart where each company's plastic output is represented by the intensity of the line's motion—higher production causes more er-

ratic vibrations.

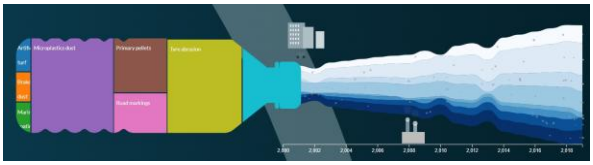


Figure 1

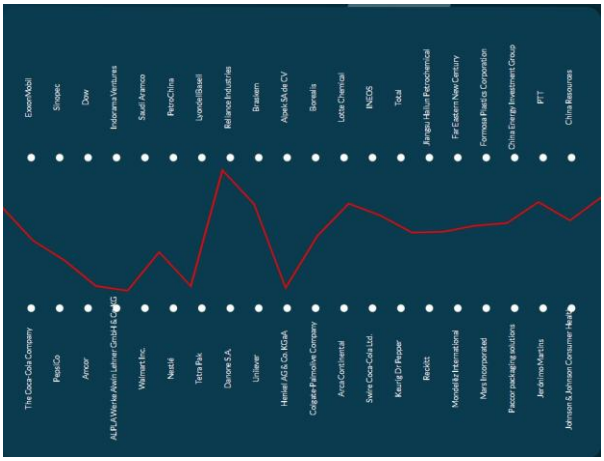


Figure 2

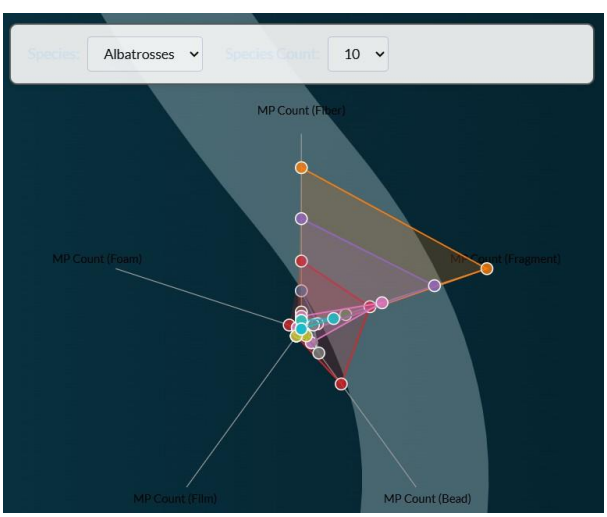
Figure 3

6 Results and Analysis

Our project leverages a series of interactive and carefully designed visualizations to uncover and communicate the lifecycle and impact of microplastics. From global sources to human exposure, each chart contributes to a compelling narrative on how microplastics originate, move, accumulate, and affect ecosystems and human health.

Treemap: Placed visually within the ”bottleneck” of our storytelling interface, the treemap offers a categorical breakdown of microplastic sources. It highlights the dominance of fibers and fragments, clearly showing their disproportionate contribution compared to films and beads. The relative sizing of each rectangle quickly informs users where interventions or regulations might be most impactful. Streamgraph: Flowing out of the bottleneck, a stream chart illustrates how microplastic concentrations have changed over time. Each band represents a geographic region, showcasing how global distribution has shifted. Notably, the stream widened significantly after 2010, indicating a global surge in microplastic presence, especially in developing regions. Bar Chart: Under the “Sources of Microplastics” section, a bar chart compares emission volumes across countries or industrial sectors. This visualization uncovers critical patterns—for example, synthetic textile industries and urban wastewater treatment plants appear as major contributors, calling attention to specific sectors in need of regulation. Donut Chart: A donut chart presents the overall proportions of different microplastic types. Fibers dominate, followed by fragments and films. This high-level view supports the need to focus mitigation efforts on fiber-related sources like laundry effluents and clothing production. Geographic Scatter Map: This map plots sampling sites worldwide where microplastics have been found. Marker size and color intensity indicate concentration levels. Clusters in coastal Asia, the Mediterranean, and urban U.S. zones underscore regional pollution patterns and possible dietary exposure risks. HeatMap: The heatmap illustrates the consumption of microplastics by humans through various dietary items in 2018. It displays the per capita average microplastic (MP) dietary intake by food type at the country level. Darker colors indicate a higher concentration of microplastics in that food. Fish stands out as the largest source of microplastics, followed by refined grains and total milk. Connected Dot Plot: This chart tracks microplastic levels found in different human organs from 2016 to 2024. Each organ is listed along the bottom, with two dots—one for each year—showing how concentrations have

changed over time. The curved lines connecting the dots help make those changes clear, whether levels have gone up or down. Larger dots mean higher concentrations, giving a quick visual sense of how microplastic exposure in our bodies has evolved. Human-Body Diagram: An anatomically inspired visualization shows microplastic exposure across various organs. Horizontal bars placed over the lungs, liver, gut, and brain quantify the relative burden. The side-by-side representation makes it easy to compare organ-level vulnerability, highlighting the gastrointestinal system and lungs as primary accumulation zones. Radar Chart: This chart compares various microplastic metrics such as counts of fibers, fragments, and films across different species. The radial layout helps identify which organisms ingest more diverse or concentrated types of plastics. Network Diagram: Finally, a brain-shaped web graph models possible interactions of microplastics with different brain regions or cellular systems. Each node and edge indicates an inferred or observed relationship. This conceptual layout supports emerging scientific theories linking neurotoxicity and cognitive effects to microplastic presence in brain tissue.



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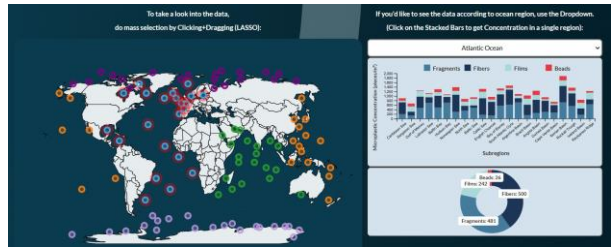

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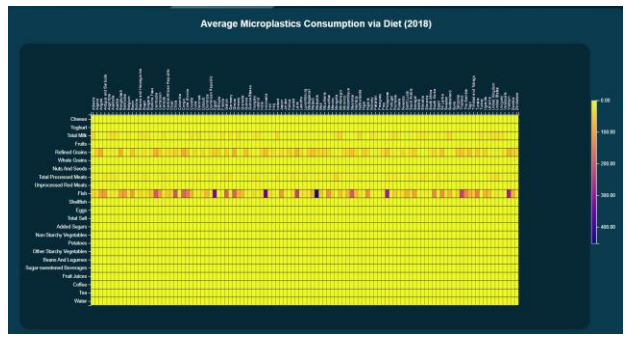

Figure 6

7 Discussion

1. Good data is hard to come by—cleaning and harmonizing disparate sources was essential.
2. Solid data underpins design; we prioritized accuracy before aesthetics.
3. Iterative teamwork refined visuals; feedback loops were critical.
4. Interactivity transformed passive charts into immersive experiences.
5. Visual metaphors (vibration) connected data to emotion.
6. Clear communication (Slack, GitHub) kept the project aligned.
7. Accessibility and responsiveness broadened audience reach.

8 Future Work

- Visualize Socioeconomic Impact Across Regions: Scope exists for charts which combine GDP data and plastic export data on a country to country basis. We do see that the countries with better developed economies tend to manage their plastic waste exports better. A chart made

with that idea could prove useful for the study of plastic exports.

- Incorporate Inclusivity by Including Accessibility Features: People with impairments like color blindness and auditory impairments would be better able to use the platform with features like voice overs, keyboard navigation, and screen-reader compatibility.
- Responsiveness: Most users engage with the internet through their mobile phones now. A wider audience will be able to engage with the project thanks to a responsive, mobile-optimized interface.

- User-Generated Data and Real-Time Mapping: An interactive map made with user generated/crowdsourced data can be cre- ated by letting the public enter data about local plastic garbage. There is every like- lihood of this map being more informative than most datasets available.

Conclusion

Our project sheds light on the pervasive and often invisible crisis of microplastic pollution. Through a combination of interactive and data-rich visualizations, we illustrated how microplastics are generated, how they travel through ecosystems, and how they ultimately infiltrate human organs. From geospatial distribution maps to organ-level concentration charts, each visualization serves to communicate not just the facts, but the urgency of this environmental challenge. The visual storytelling approach enabled us to move beyond static facts and figures, allowing the audience to explore, engage, and internalize the scale and consequences of microplastic pollution. It is our hope that this work sparks deeper awareness, drives behavioral change, and inspires further research into sustainable alternatives and policy reforms.

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